Math 141, S10 **Name:** Answer

Exam 1 **Section:** **SSN: xxx-xx-**

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	18	
2	18	
3	24	
4	24	
5	16	
Total	100	

Good Luck !

Problem 1.(18 points) Find the natural domain of each of the following functions.

(a) $f(x) = \dfrac{1}{x^2 - 9}$

$x^2 - 9 \neq 0 \Rightarrow x^2 \neq 9$

$\Rightarrow x \neq 3, -3$

domain: $(-\infty, -3) \cup (-3, 3) \cup (3, +\infty)$

(b) $g(x) = \sqrt{4x - x^2}$

$4x - x^2 \geqslant 0$

$\Rightarrow x(4-x) \geqslant 0$

$\Rightarrow 0 \leqslant x \leqslant 4$

domain: $[0, 4]$

(c) $h(x) = \dfrac{1}{1 - e^x}$

$1 - e^x \neq 0$

$\Rightarrow e^x \neq 1$

$\Rightarrow x \neq 0$

domain: $(-\infty, 0) \cup (0, +\infty)$

Problem 2. (18 points)

(a) Let $f(x) = \dfrac{x}{1-x}$ and $g(x) = \cos x$, find the composite functions $f \circ g$ and $g \circ f$.

$$f \circ g(x) = f(g(x)) = \frac{\cos x}{1-\cos x}$$

$$g \circ f(x) = g(f(x)) = \cos\left(\frac{x}{1-x}\right)$$

(b) Find a formula for the inverse of the function $f(x) = \dfrac{4x-1}{2x+3}$.

$$y = \frac{4x-1}{2x+3}$$

$$y(2x+3) = 4x-1$$

$$2xy+3y = 4x-1$$

$$2xy - 4x = -3y-1$$

$$(2y-4)x = -3y-1$$

$$\Rightarrow \quad x = \frac{-3y-1}{2y-4} \qquad \Rightarrow f^{-1}(x) = \frac{-3x-1}{2x-4}$$

(c) Simplify

$$3^2 \cdot 3^{-1} + \log_4 2 + \log_4 8 + e^{\frac{1}{2}\ln 9}$$

$$= 3^{2-1} + \log_4(2\cdot 8) + e^{\ln 9^{\frac{1}{2}}}$$

$$= 3 + \log_4 16 + e^{\ln 3}$$

$$= 3+2+3$$

$$= 8$$

Problem 3. (24 points) Find the following limits.

(a) $\lim_{x\to 1} e^{x^3-x}$

$$= e^{\lim_{x\to 1} x^3-x}$$

$$= e^{1^3-1} = e^0 = 1$$

(b) $\displaystyle\lim_{x\to -2} \frac{x^2-4}{x^2-x-6}$

$$= \lim_{x\to -2} \frac{\cancel{(x+2)}(x-2)}{\cancel{(x+2)}(x-3)}$$

$$= \frac{-2-2}{-2-3}$$

$$= \frac{4}{5}$$

(c) $\displaystyle\lim_{x\to +\infty} \frac{x^3-1}{2x^3+x^2-10}$

$$= \lim_{x\to +\infty} \frac{\frac{x^3-1}{x^3}}{\frac{2x^3+x^2-10}{x^3}}$$

$$= \lim_{x\to +\infty} \frac{1-\frac{1}{x^3}}{2+\frac{1}{x}-\frac{10}{x^3}}$$

$$= \frac{1}{2}$$

(d) $\displaystyle\lim_{x\to 0^-} \frac{1}{\sin x}$

$$= -\infty$$

Problem 4. (24 points)
(a) Explain why this function is discontinuous at $x = 0$:

$$f(x) = \begin{cases} \dfrac{x^2 - 2x}{|x|} & x \neq 0, \\ -2 & x = 0. \end{cases}$$

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} \frac{x^2 - 2x}{x} = \lim_{x \to 0^+} \frac{x(x-2)}{x} = -2$$

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} \frac{x^2 - 2x}{(-x)} = \lim_{x \to 0^-} -(x-2) = 2 \neq \lim_{x \to 0^+} f(x)$$

$\Rightarrow f$ is discontinuous at $x = 0$.

(b) Let $f(x) = x^3 - x$. Find $f'(x)$ and $f''(x)$.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{[(x+h)^3 - (x+h)] - [x^3 - x]}{h}$$

$$= \lim_{h \to 0} \frac{[\cancel{x^3} + 3x^2h + 3xh^2 + h^3 - \cancel{x} - h] - [\cancel{x^3} - \cancel{x}]}{h}$$

$$= \lim_{h \to 0} \frac{3x^2h + 3xh^2 + h^3 - h}{h}$$

$$= \lim_{h \to 0} (3x^2 + 3xh + h^2 - 1)$$

$$= 3x^2 - 1$$

$$f''(x) = \lim_{h \to 0} \frac{f'(x+h) - f'(x)}{h} = \lim_{h \to 0} \frac{[3(x+h)^2 - 1] - [3x^2 - 1]}{h}$$

$$= \lim_{h \to 0} \frac{[\cancel{3x^2} + 6xh + 3h^2 - \cancel{1}] - [\cancel{3x^2} - \cancel{1}]}{h}$$

$$= \lim_{h \to 0} \frac{6xh + 3h^2}{h}$$

$$= \lim_{h \to 0} (6x + 3h)$$

$$= 6x$$

Problem 5. (16 points) Let $f(x) = \dfrac{2}{3-2x}$.

(a) Find $f'(x)$.

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

$$= \lim_{h\to 0} \frac{\frac{2}{3-2(x+h)} - \frac{2}{3-2x}}{h}$$

$$= \lim_{h\to 0} \frac{\frac{2}{3-2x-2h} - \frac{2}{3-2x}}{h}$$

$$= \lim_{h\to 0} \frac{\frac{2(3-2x)-2(3-2x-2h)}{(3-2x-2h)(3-2x)}}{h}$$

$$= \lim_{h\to 0} \frac{(\cancel{6}\cancel{-4x})-(\cancel{6}\cancel{-4x}-4h)}{(3-2x-2h)(3-2x)h}$$

$$= \lim_{h\to 0} \frac{4\cancel{h}}{(3-2x-2h)(3-2x)\cancel{h}}$$

$$= \frac{4}{(3-2x)^2}$$

(b) Use the result of (a) to find an equation of the tangent line to the curve $y = \dfrac{2}{3-2x}$ at the point $(1, 2)$.

$$f'(1) = \frac{4}{(3-2\cdot 1)^2} = \frac{4}{1^2} = 4.$$

$$\Rightarrow \quad \frac{y-2}{x-1} = f'(1)$$

$$\frac{y-2}{x-1} = 4$$

$$y-2 = 4(x-1)$$

$$4x - y - 2 = 0$$